How Often are Ethical Principles Discussed in Human-in-the-Loop Machine Learning Papers? - A Systematic Literature Review

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***Abstract*—Human-in-the-Loop (HITL) techniques are increasingly being explored in applications related to Machine Learning (ML). Simultaneous to the power of ML models increasing, concerns over the ethical implications that AI may bring to society have also increased. A broader interest in ethics including from legal and social scholars outside the field of computer science has led to enumeration of ethical principles in previous studies. Here we present a systematic literature review of the discussion of ethical principles in studies related to HITL ML in technical Computer Science papers. We find meaningful discussion of ethical principles occurs in only 26% of papers and often focuses on specific principles in relation to specific subdomains.**

I. Introduction

Recent advancements in Machine Learning (ML) have led to the development of sophisticated ML systems, yet many tasks still require human involvement due to the unique capabilities humans offer, which has led to an increasing tendency of integrating human input into machine learning processes and decision-making. [1] In the area of ML, the integration of Human-in-the-Loop (HITL) systems introduces significant ethical challenges. [2] These challenges encompass issues related to accountability, fairness, and transparency. However, there exists a gap in understanding how software engineers assess and address these ethical implications within the context of HITL in ML research literature. The focus on ethics within software engineering reflects the increasing significance of software-driven systems in society, prompting developers to acknowledge and address the potential ethical issues of their creations, given their impact on individuals and societal values. [3] This project aims to research how Software Engineers Assess the Ethical Impacts of Human-in-the-Loop in Machine Learning Research Literature by conducting a systematic literature review to identify and analyze the ethical concerns associated with HITL in ML technologies. By doing so, we seek to develop a set of ethical guidelines and assessment frameworks for HITL in ML systems, specifically tailored for software engineers.

While the application of HITL in ML systems holds immense potential for enhancing algorithmic performance and addressing complex real-world problems, it also introduces significant ethical challenges for software engineers [2]. As human judgment and intervention are integrated into the ML pipeline, questions arise regarding who should be held accountable for the decisions made by these systems. Traditional notions of accountability in software engineering may not fully encompass the complexities introduced by human involvement, leading to challenges in assigning responsibility for errors, biases, or adverse outcomes. Despite the presence of codes of ethics and ethical principles for software professionals, software engineers encounter ongoing difficulties in navigating ethical dilemmas and making ethically sound decisions. [4] Integrating HITL in ML presents ethical challenges, with a notable gap in how software engineers evaluate and address these issues to uphold accountability, fairness, and transparency.

II. Proposed Solution

There are few systematic literature review related to this topic that we have found so far. [5], [6] We aim to conduct a systematic literature review to comprehensively analyze existing research literature on HITL in ML technologies. By synthesizing and evaluating the findings from these studies, clear ethical guidelines and assessment frameworks will be developed specifically for software engineers. The proposed guidelines will offer software engineers a structured approach to assessing and mitigating ethical risks in HITL-based ML systems

and will also serve as a road map for navigating the ethical complexities of integrating human involvement into ML systems, providing practical strategies and recommendations for addressing ethical concerns effectively.

III. Research Questions

Addressing these challenges requires focused research. Therefore, following the problem statement, we outline our research questions which will guide our systematic literature review and analysis:

RQ 1 How frequently are ethical principles mentioned in the context of HITL ML literature?

RQ 2 Which ethical principles of HITL in ML technologies are described in the technical literature?

RQ 3 In what applications are these ethical principles being applied?

The answers to these questions will not only help in understanding the current state of ethical considerations in HITL systems but also contribute to developing a set of ethical guidelines and frameworks tailored for software engineers working with HITL in ML.

IV. Related Works

While we have searched for other papers reviewing ethical practices surrounding HITL systems for machine learning, we have not uncovered a comprehensive discussion or review focused on the ethics of HITL practices in the software engineering community. Nor have we found uniform discussion among papers that are more focused on narrower aspects of HITL for machine learning practices. It is not obvious to us that their exist common ethical frameworks, principles or approaches that software engineers or researchers apply to the topic.

Multiple previous studies have touched on aspects of HITL in ML systems. Following is a profile of some of these. Our intention is to conceptually build from these reviews and studies to better understand our own paper's focus.

A. HITL Review Literature

Previous reviews of HITL have focused on the overall technical state of the field and enumerating current challenges [2][1][7]. In particular Wang et al. explore the level of involvement humans have at every level of the machine learning process.[1]

Mosqueira et al. expends significantly more effort classifying the level and type of human involvement at different stages in the machine learning pipeline.[7] Especially of note are HITL based strategies intended to add human oversight over a machine learning process. These purportedly improve the ethical and/or technical characteristics of a system. Also discussed are the impacts of the technology on humans who may participate directly in the HITL system. Furthermore, this paper denotes categories of HITL processes that merit further investigation including Explainable AI, Usable AI, and Useful AI. These categories of machine learning focus on systems that explainable to humans. They don't just provide an answer or outcome, but explain why that's the correct answer or how the answer was arrived at.

There is also a substantial amount of literature addressing HITL machine learning techniques in medical applications or other health science contexts.[3][8] Obviously there are a multitude of concerns with these systems including security, privacy, and possible life or death decisions or consequences.

B. Ethics in Computer Science and AI

Ethics and ethical concerns in software engineering have been covered broadly before. [9][4][5][10][11][12]

Gogoll et al. discuss ethical Codes of Conduct employed in both business and academic environments and contrasts their effectiveness with ethical deliberations. In particular attention is paid to delineate the ethical concerns appropriate for software engineers to consider as opposed to lawyers or business managers. They attempt to distinguish and critique the use of ethical Codes of Conduct from ethical deliberation practices and evaluate the practicality and effectiveness of each.[13]

With more recent advances in AI technology, increasing scrutiny has followed. Specific ethical concerns with AI systems, including machine learning, have accelerated over the past decade as AI systems increase in commercial viability and technological advancement. Khan et. al. presents a recent systematic literature review of ethics in AI to survey principles and challenges across that domain.[6] Morley et al. in particular attempt to describe best ethical practices in Machine Learning system development as opposed to simply noting ethical concerns.[14]

V. Methods

We chose to carry out a systematic literature review to answer our research questions. In order to achieve objective results the study was carried out according to standard literature review practices.[15]

A. Data Source

We started by surveying the ACM Digital Library as our data source. We are interested in holding this

literature repository as representative of papers written by and for the software engineering and development community as opposed to reference material authored or intended for other audiences (such as the legal or business communities). Our project is focused on understanding, methodologies, and practices of the software engineering and research community itself. We are also limited by time from incorporating additional literature repositories.

B. Search String

Assembly of the search terms and strings for this study required some nuance. Our goal was to find studies discussing both HITL and ML. Different authors use different abbreviations and hyphenations for HITL such as "HITL" (our preferred abbreviation), "HIL", "human in the loop" (no hyphenation), etc. To overcome this variation in terminology, we instead search for both "human" and "loop" to be included in the text. To find papers related to machine learning we require inclusion of the strings "machine learning" or "ML". We decided against searching for specific ethical principles because we were interested in finding papers without discussions of any specific principles as well.

The terms HITL and ML are used often in computer science literature. Performing a full-text search returned far too many papers to manually review. Therefore we opted to require our search terms be included in abstracts. Our operating assumption is that authors are more likely to include these terms in the abstract if those topics were more relevant to the paper rather than a fleeting reference in the full-body text.

We also decided to require our papers be published before 2024 (our current year). This was primarily to make searches in the ACM Digital Library reproducible while writing the study, and to exclude data from publications in process or incomplete statistics for a partial year (2024).

The resulting search string used to search the ACM Digital Library is presented in Table I for reference. This search returned 192 items from ACM's online database which were used in following steps.

TABLE I: Search Query String

ACM Digital Library Search Query
"query": { Abstract:(human) AND Abstract:(loop) AND Abstract:("machine learning" OR "ML") } "filter": { E-Publication Date: (* TO 12/31/2023), ACM Content: DL }

C. Initial Screening

Bibliographic data for the items returned in the previous query were downloaded and converted into an excel spreadsheet. This spreadsheet was then used to manually screen each study for further inclusion in the study by one of the authors. Reasons for exclusion were marked for any excluded papers, and the spreadsheet is available in this paper's GitHub repository.

Exclusion criteria included:

- Body of research item was not primarily a textual presentation of research. Conference proceedings prefaces or summaries were excluded. Similarly abstracts from keynotes, invited talks, industry-day talks, tutorials, etc. were discarded from further examination.
- Full text of item did not appear to be available or linked through the ACM digital library. (A notable problem with several thesis dissertations.)
- Item's abstract did not include keywords from the search query, but was mistakenly returned. (Possibly this represents a glitch in the ACM Digital Library search functionality.)
- Item's abstract did contain all relevant keywords, but upon closer examination, work was clearly not substantively related to both HITL and ML topics.

Inclusion criteria included:

- Must be published prior to 2024.
- Must be written in English.
- Must appear as an entry in the ACM Digital Library.

All items met the inclusion criteria. However, 118 items met exclusion criteria. These were removed from the literature pool leaving 74 papers for further review and data extraction.

D. Data Extraction

Data extraction occurred primarily in three categories: basic bibliographic/publication information, assessment of discussion of ethical principles, and categorization of HITL ML learning model.

To assess the inclusion of ethical principles in a particular study, a full text search of each paper was performed for ethical keywords drawn from the work of Khan et al. [6]. We attempt here to extend their work by using the list of ethical principles they found relevant to characterizing AI research. If keywords were found in a specific paper, the context and use of the keywords was examined in detail by the authors to determine whether the surrounding discussion addressed or otherwise described consideration of ethical issues. This

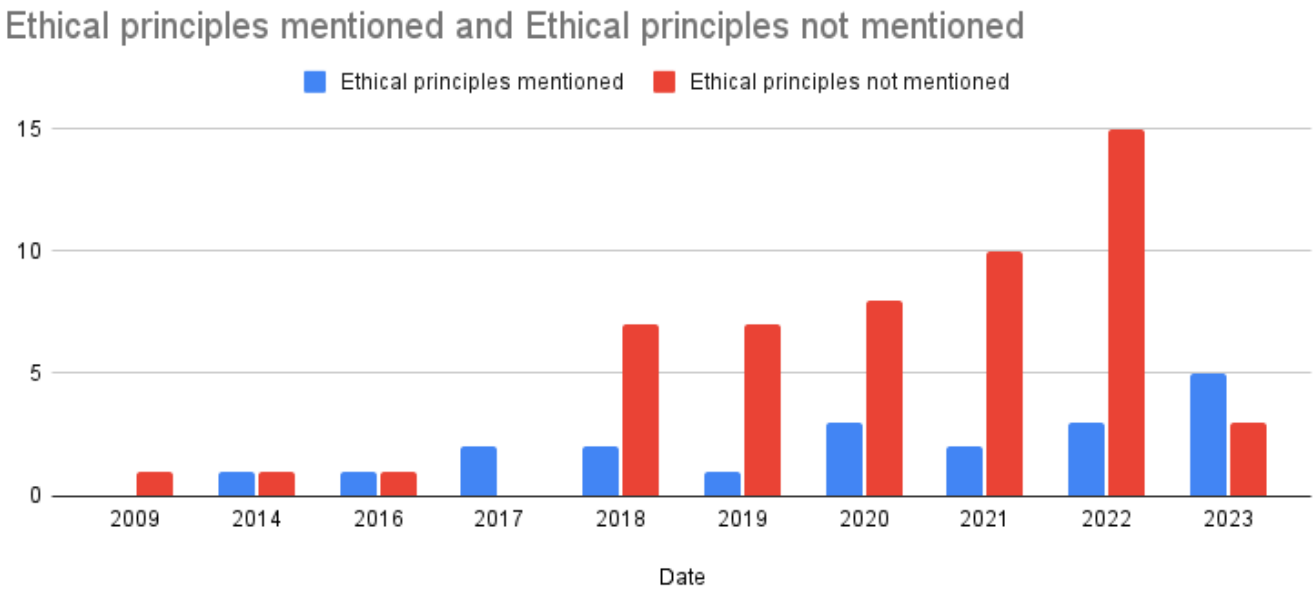

Fig. 1: Annual quantity of publications with and without a discussion of an ethical principle from 2009 to 2023

manual review was intended to prevent mis-classification in case the keywords describing ethical principles were reused outside the context of an ethical discussion. In considering whether ethical discussions occurred, we discounted mentions of human institutional review boards as part of routine interview work, etc. Our interest in this study is not to judge ethical research conduct of authors, but to focus on the broader ethical topics discussed and the ethical ramifications of specific tools or application domains.

Further, we also searched the papers for discussion of specific HITL ML learning models. As per the work of Mosqueira-Rey et al. [7] we decided to attempt categorization of papers according to whether they included discussions of Active Learning, Interactive Machine Learning, or Machine Teaching. This categorization too was carried out first using full-text keyword searches for these learning techniques followed by manual review to check appropriate contextual meaning.

All data was gathered in a google spreadsheet from which we draw to make further tables included here. This spreadsheet is also available in this project's GitHub repository.

VI. Results

A. RQ1: How frequently are ethical principles discussed?

Out of the 74 items reviewed in the data extraction step, 54 did not include any substantive reference to ethical concerns or principles while 19 did. Figure 1 shows the distribution of these papers according to year of publication.

Production of papers discussing ethical principles related in HITL ML has been relatively modest. The oldest papers returned in our search were from 2014. Following a two year gap, we note a steady drip of publications have including some form of ethics discussions from 2018 onward.

However, overall publications discussing both HITL and ML have increased every year since 2018 with the exception of 2023. So discussion of ethical principles in relevant papers seems to have decreased as a percentage during the same time period.

It should be noted that the overall number of publications examined in our study is low and the nature of the papers is highly heterogeneous. While Figure 1 may reflect broad trends, we think it's inappropriate to draw detailed statistical conclusions using this dataset.

B. RQ2: Which ethical principles are being discussed?

Table II reflects the number of times an item from our data extraction collection was flagged as including

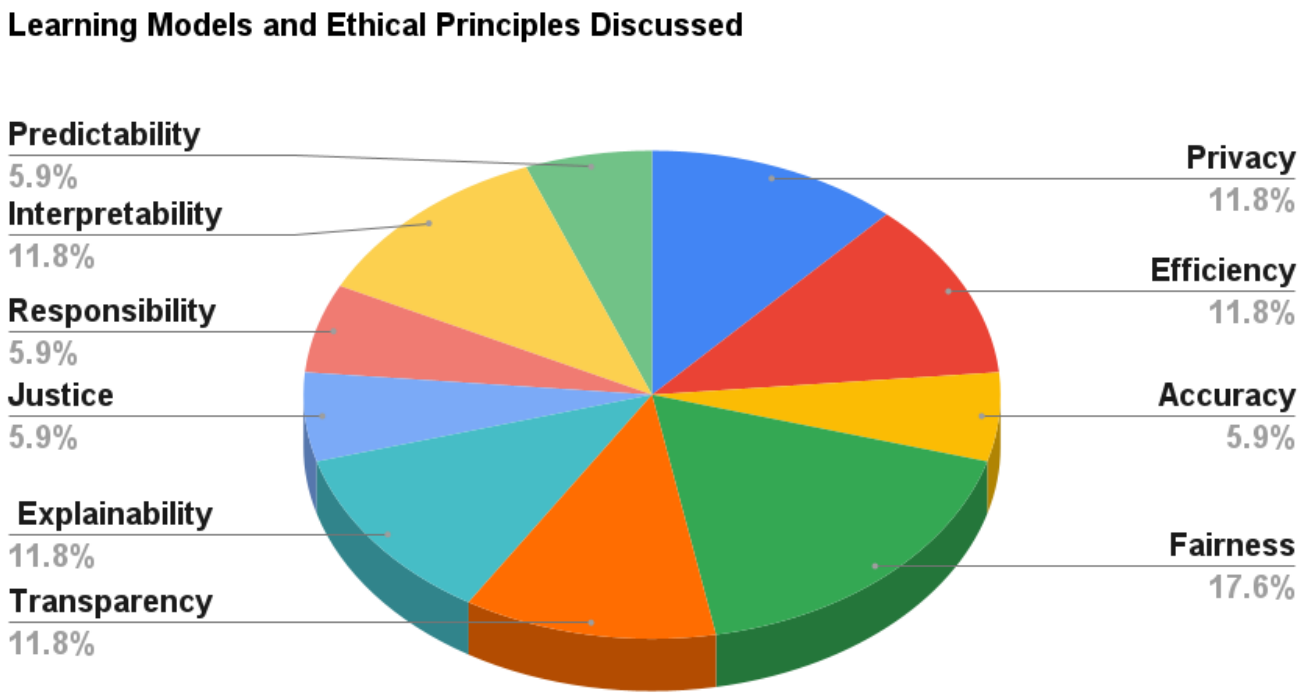

Fig. 2: Percentage of Ethics Principles discussed among literature sample

TABLE II: Count of Papers Discussing each Ethical Principle

Ethical Principle	# of Papers
Privacy	4
Efficiency	1
Trust	1
Transparency	8
Accountability	3
Fairness	8
Explainable	7
Justice	7
Accuracy	4
Interpretability	8
Sustainability	1
Responsibility	6
Effectiveness	2
Predictability	2
Freedom	1

a substantial discussion related to a specific ethical principle keyword. Figure 2 displays how commonly ethical principles are referenced as a percentage of the total count of ethical principle references throughout our literature sample.

The principles discussed most often were Explainability, Fairness, Interpretability, Justice, Responsibility, Transparency. Each of these were discussed in between 6 and 8 studies.

The principles of Accountability, Accuracy, Effectiveness, Freedom, Predictability, Privacy, Sustainability appeared in between 1 and 4 of the studies.

We did not observe any instances of references to the terminology of Autonomy, Beneficence, Data Security, Human Dignity, Non-maleficence, Prosperity, Safety, Solidarity.

Table IV shows how often two specific ethical principles are mentioned in the same paper. We include only principles mentioned at least once within one of the examined studies in this table for brevity. We note that papers discussing at least one ethical principle are often discussing at least one other as well.

C. RQ3: What applications are these ethical principles being applied?

Applications of the ethical principles discussed in papers were sometimes difficult to discern. Some papers dealt with a narrow application of HITL or ML to a specific use-case or domain. Other papers attempted to

Fig. 3: Word cloud of domains and applications where human-in-the-loop (HITL) was researched

TABLE III: Count of Papers mentioning a specific HITL ML category

Learning Techniques	Paper Count
Active Learning	10
Interactive Machine Learning	22
Machine Teaching	1
Total:	33

create broadly applicable tools that could be extended or applied to multiple problem domains. In particular, some papers might discuss an example application, but also suggest broader applicability of their tools, processes, or results.

Figure 3 shows a word cloud generated of various application and domain keywords found throughout the papers.

D. *What types of HITL ML are discussed and how often?*

After beginning our review we also decided to gather information about one further area of interest. Some experts have suggested that HITL ML techniques may be categorized into specific techniques mentioned earlier in the paper such as Active Learning, Interactive Machine Learning, and Machine Teaching. (cite reference) We attempted to categorize papers in our study according to whether they studied or proposed solutions based primarily around one of these techniques.

Table III shows the data categorization data we gathered from our literature sample. Out of the 74 papers examined, 33 appeared to substantially discuss one of these techniques. The primary technique of interest to authors was Interactive Machine Learning at two thirds of papers discussing one of these HITL ML techniques.

Figure 4 shows which ethical principles are discussed among papers presenting an identifiable HITL learning technique. Interestingly, ethical principles are almost always discussed in association with Interactive Machine Learning.

VII. Discussion

Most of the studies in our review did not discuss a HITL ML technique from the categories laid out in the work of Mosqueira-Rey et al. While humans were involved in all these studies their roles were often outside involvement with machine learning training. In some papers, humans were described as part of the loop when they received information from a ML model or just provided it with a request. [16] [17] [18] [19] Other papers discussed ML but the Human in the loop was involved with a different component of the technology stack. [20] [21] [22]

One particular topic of study that flips the idea of HITL ML on its head is referred to as “Algorithm-in-the-Loop”. [23] [24] This terminology appears to emphasize the study of human decision making processes where an algorithm plays an important role. It is not clear that this terminology necessarily implies HITL involvement in training the relevant ML models, but these papers may be of interest for similar ethical reasons. These studies were coincidentally included in our review because they included the keywords of “human” and “loop” in their abstracts, but may represent an important direction pool of papers not fully explored by this study.

Most of the papers discussing ethics were clustered around similar and related principles of fairness, transparency, explainability, and interpretability. These papers explored the application of ML models to help humans assess and classify other humans behaviors or characteristics [25] [26] [27] [28] [20]. These model are then used for purposes of judging humans or making decisions that may have direct moral impacts on them [29] [30] [24] [23] [31]. Intuitively this makes sense: the most important assessment of ethical principles will in fact concern their application and impact on humans themselves.

Interestingly, the suggested and studies role of HITL across these different papers were not uniform. Some

TABLE IV: Overlap in Discussion of Ethical Principles Within Papers

Ethical Principle	Privacy	Efficiency	Trust	Transparency	Accountability	Fairness	Explainable	Justice	Accuracy	Interpretability	Sustainability	Responsibility	Effectiveness	Predictability	Freedom
Privacy				1		1	1	1		1		1			
Efficiency		1													
Trust															
Transparency					2	4	5	4	2	4	1	2	1	1	1
Accountability							2	1		1	1	1	1	1	
Fairness							3	4	3	3		2	1	1	1
Explainable								2	2	3			1	1	
Justice									1	2					
Accuracy										1			1	1	
Interpretability												2			
Sustainability															
Responsibility															
Effectiveness														1	
Predictability															
Freedom															

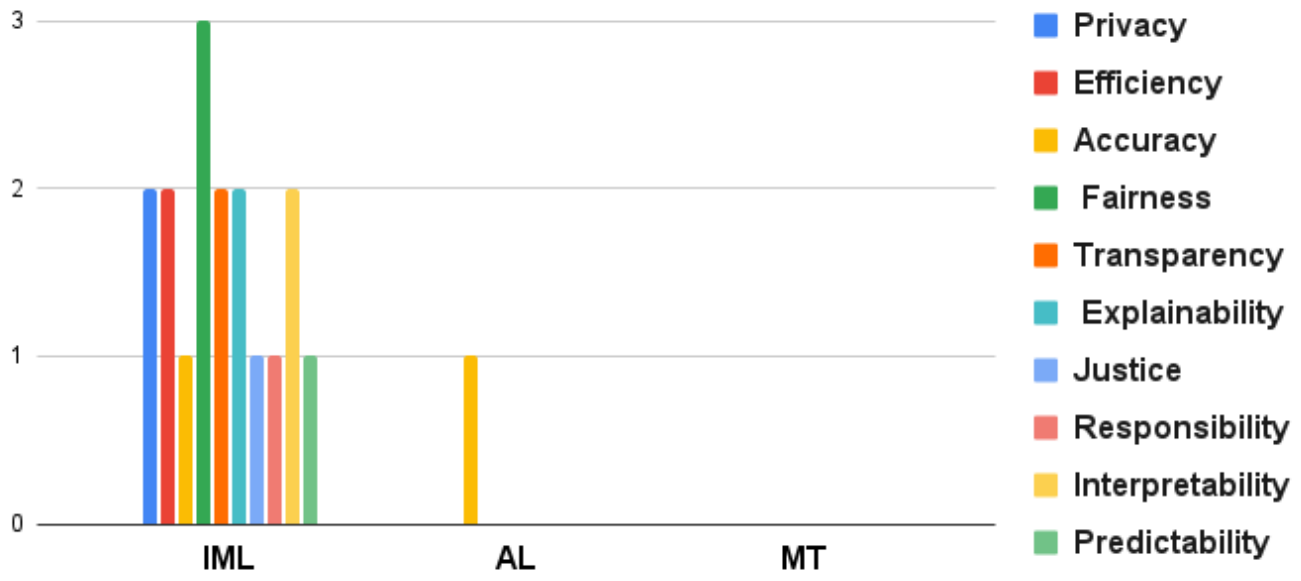

Fig. 4: Ethical Principles discussed in relation to HITL ML Learning Models

papers focused on measuring and mitigating human bias in their models. Effectively the role of humans was minimized and their input treated skeptically as in the work of Zamfirescu-Pereira et al. [26] Other papers treated human input as intrinsically necessary to create ML models that could be shown and explained as reasonably fair such as the work of Zhang et al. [25] The latter study in fact suggests building a HITL ML system, so that humans can decide what is considered fair and why.

Overall, some of these studies begin to bridge the gap between principles and practical application of ethical best-practices noted in other works [14]. HITL ML models may play a role in this. But the number of papers we found in this review with relevant overlap between HITL ML and discussion of ethical principles is quite small and many have been published quite recently. It does not seem appropriate to draw meaningful conclusions about this area of the literature at this time.

VIII. Limitations and Challenges

The authors of this paper have found that discussions of ethical principles in computer science studies are difficult to characterize. Often authors appear primarily interested in the technical details and performance metrics that characterize HITL ML technologies and solutions. With notable exceptions for papers that attempt to address issues of fairness, interpretability, and fairness head on, most papers appeared to be obliquely interested in the ethical impacts their work may have. This makes it almost intrinsically subjective to judge much of the limited ethical discussion authors often choose to include in their papers.

A substantial challenge and limitation to this work is a result of the non-uniformity of terminology. Authors do not necessarily agree on basic definitions or usage of terminology in regards to ethical topics or classification of HITL ML training techniques. The very recent work Khan et al. [6] appears to be one of the first papers to attempt disentangling and clarifying much of the terminology and discussion that's evolved organically in literature over longer than the last 10 years. With so much room for different definitions and interpretation, careful classification of papers in the categories we have chosen was difficult at best and likely included subjectivity on our part.

It was nearly impractical to assess the inclusion of Accuracy as an ethical principle in papers. Many studies discuss and reference the Accuracy of their ML models, but it's generally difficult to disentangle whether this discussion is related to meaningful ethical concerns rather than practical model performance metrics. We therefore discarded consideration of this keyword during the review.

Another prime challenge in this work required choosing a representative set of papers to analyze. Our criteria of searching for relevant keywords in abstracts was intended to prioritize the most relevant papers. Because

different authors may view the terminology of HITL more or less important to the greater context of their work, these keywords may not appear in the abstracts of many papers which we would otherwise consider relevant to our literature review.

Finally, the work of Khan et al. was not intended to generate searchable keywords with which to characterize other papers. Their work was instead intended to gather conceptual principles from a cross section of papers focused on ethical concerns. Papers in our literature review may have touched on these concerns, but used different keywords or described concepts with other terminology.

Many papers referenced ethical concerns in their related works, but did not seem to discuss them directly in their own paper. We decided that while these may have been motivating or contextual concerns for a given paper, they did not rise to the level of an ethical issue discussion if the author's own work did not expand on or directly address them in a meaningful way. We grant this is another area of subjectivity and personal judgement.

Finally, we were limited on time. Their were a substantial number of papers and it was difficult to choose appropriate metrics and methodology that were relevant to the time period when this work required completion.

IX. Future Work

The findings from this systematic literature review open several pathways for future research in the ethical implications of Human-in-the-Loop Machine Learning systems.

One immediate area for further exploration is the expansion of the search scope within the literature. Future studies could benefit from applying the search queries used in this review not only to the abstracts but also to the full text of articles. This approach may uncover a more comprehensive array of discussions on ethical issues that were not prominently featured in the abstracts but were discussed in the body of the articles.

Additionally, while our research questions have provided foundational insights into the current state of ethical considerations in HITL ML, there is a significant opportunity to refine these questions for a deeper understanding. Future researchers could re-frame these questions to more specific ethical dimensions more thoroughly or to explore ethical considerations across various domains systematically. For instance, questions could be aimed at understanding how ethical principles are implemented in practice within different sectors such as healthcare, automotive, or finance, which might have unique ethical challenges and solutions. Moreover, subsequent reviews could explore the variability in ethical emphasis across different domains of application. This approach would not only show how ethical principles are prioritized depending on the application area but also could guide sector-specific ethical guidelines and frameworks.

Furthermore, future research could benefit from adopting a longitudinal approach, tracking how the integration of ethical considerations in HITL ML evolves over time as technologies develop. This analysis would offer more temporal understanding of the shifts in ethical practices, providing a clearer picture of the trajectory of ethical considerations in technology deployment.

By extending the methodologies and questions of this review, future work can continue to advance our understanding of the ethical landscapes in HITL ML systems.

X. Conclusion

In the conclusion of our systematic literature review, it is evident that ethical considerations are still not uniformly emphasized in the field of Human-in-the-Loop Machine Learning (HITL ML), with ethical principles explicitly mentioned in only approximately 26 percent of the surveyed literature. Our first research question revealed this lack of frequency, underscoring a significant gap in the ethical discourse within technical documentation.

In response to our second question, it became clear that when ethical principles are discussed, transparency, fairness, and interpretability are the most frequently cited concerns, indicating a community awareness of these critical issues, despite their infrequent mention overall.

Finally, addressing our third research question, we discovered that the application of HITL ML technologies spans a broad range of environments, with the educational and research sectors emerging as the foremost areas of implementation. This prevalence suggests a growing recognition of the importance of ethical considerations in domains where the impact on human learning and knowledge creation is most direct and potentially profound. Moving forward, this insights calls for a concerted effort to integrate ethical discussions more deeply and consistently across all sectors utilizing HITL ML, not only to enhance the credibility and acceptance of ML systems but also to align them more closely with societal values and ethical standards.

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